Statistical Testing

Nonparametric

chi-square test

Parametric

ANOVA

chi-square test

A Pearson's **chi-square test** is a statistical for categorical data .It is used to determine whether your **data are significantly different** from what you **expected**. Chi-square is often written as $\mathbf{X^2}$ and is pronounced "kai-square" (rhymes with "eye-square"). It is also called chi-squared.

chi-square tests, are among the most common **nonparametric tests**.
Nonparametric tests are used for data that don't follow
the assumptions of parametric tests , especially the
assumptions of a normal distribution.

There are two types of Pearson’s chi-square tests:

chi-square goodness of fit test

It is used to test whether the frequency distribution of a categorical variable is different from your expectations.

chi-square test of independence

It is used to test whether two categorical variables are related to each other.

Example: chi-square goodness of fit test

Example: Bird species at a bird feeder

Frequency of visits by bird species at a bird feeder during a 24-hour period

Bird species	Frequency
House sparrow	15
House finch	12
Black-capped chickadee	9
Common grackle	8
European starling	8
Mourning dove	6

A chi-square test (a chi-square goodness of fit test) can test whether these observed frequencies are significantly different from what was expected, **such as equal frequencies.**

Example: chi-square test of independence

Example: Handedness and nationality

Contingency table of the handedness of a sample of Americans and Canadians

	Right-handed	Left-handed
American	236	19
Canadian	157	16

A chi-square test (a test of independence) can test whether these observed frequencies are significantly different from the frequencies expected **if handedness is unrelated to nationality.**

chi-square

$$X^2 = \sum \frac{(O-E)^2}{E}$$

Where X^2 is the Chi-Square test symbol

Σ is the summation of observations

O is the observed results

E is the expected results

Example

Let's say you want to know if gender has anything to do with political party preference. You poll 440 voters in a simple random sample to find out which political party they prefer. The results of the survey are shown in the table below:

	Republican	Democrat	Independent	Total
Male	100	70	30	200
Female	140	60	20	220
Total	240	130	50	440

Step 1: Define the Hypothesis

- H0: There is no link between gender and political party preference.
- H1: There is a link between gender and political party preference.

$$\text{Expected Value} = \frac{(Row\ Total) * (Column\ Total)}{Total\ Number\ Of\ Observations}$$

For example, the expected value for Male Republicans is:

$$= \frac{(240) * (200)}{440} = 109$$

	Republican	Democrat	Independent	Total
Male	100	70	30	200
Female	140	60	20	220
Total	240	130	50	440

Step 2: Calculate the Expected Values

$$a_{ij} \xrightarrow{\text{Rows (m)}}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \Big\downarrow \text{Columns (n)}$$

	Republican	Democrat	Independent	Total
Male	100	70	30	200
Female	140	60	20	220
Total	240	130	50	440

Observed Values

Expected Values

	Republican	Democrat	Independent	Total
Male	109	59	22.72	200
Female	120	65	25	220
Total	240	130	50	440

Expected Values

Step 3: Calculate (O-E)2 / E for Each Cell in the Table

Now you will calculate the (O - E)2 / E for each cell in the table.

Where

O = Observed Value

E = Expected Value

$(O-E)^2/E$				
	Republican	Democrat	Independent	Total
Male	0.74311927	2.050847	2.332676056	200
Female	3.33333333	0.384615	1	220
Total	240	130	50	440

Step 4: Calculate the Test Statistic X2

X2 is the sum of all the values in the last table

= 0.743 + 2.05 + 2.33 + 3.33 + 0.384 + 1

= 9.837

(total rows-1)(columns-1) (2-1)(3-1) = 2

Critical values of the Chi-square distribution with *d* degrees of freedom

Probability of exceeding the critical value

d	0.05	0.01	0.001	*d*	0.05	0.01	0.001
1	3.841	6.635	10.828	11	19.675	24.725	31.264
2	5.991	9.210	13.816	12	21.026	26.217	32.910
3	7.815	11.345	16.266	13	22.362	27.688	34.528
4	9.488	13.277	18.467	14	23.685	29.141	36.123
5	11.070	15.086	20.515	15	24.996	30.578	37.697
6	12.592	16.812	22.458	16	26.296	32.000	39.252
7	14.067	18.475	24.322	17	27.587	33.409	40.790
8	15.507	20.090	26.125	18	28.869	34.805	42.312
9	16.919	21.666	27.877	19	30.144	36.191	43.820
10	18.307	23.209	29.588	20	31.410	37.566	45.315

INTRODUCTION TO POPULATION GENETICS, Table D.1

X2 is the sum of all the values in the last table

= 0.743 + 2.05 + 2.33 + 3.33 + 0.384 + 1

= 9.837

ANOVA test (Analysis Of Variance)

Score	Frequency
6	2
7	3
8	7
9	7
10	1

Sample 1

Score, x	Frequency, f
2	2
3	5
4	11
5	7
6	10
7	9
8	6

Sample 2

Score	Frequency
5	2
6	3
7	2
8	2
9	1
10	1

Sample 3

Why to check variance but not mean ?

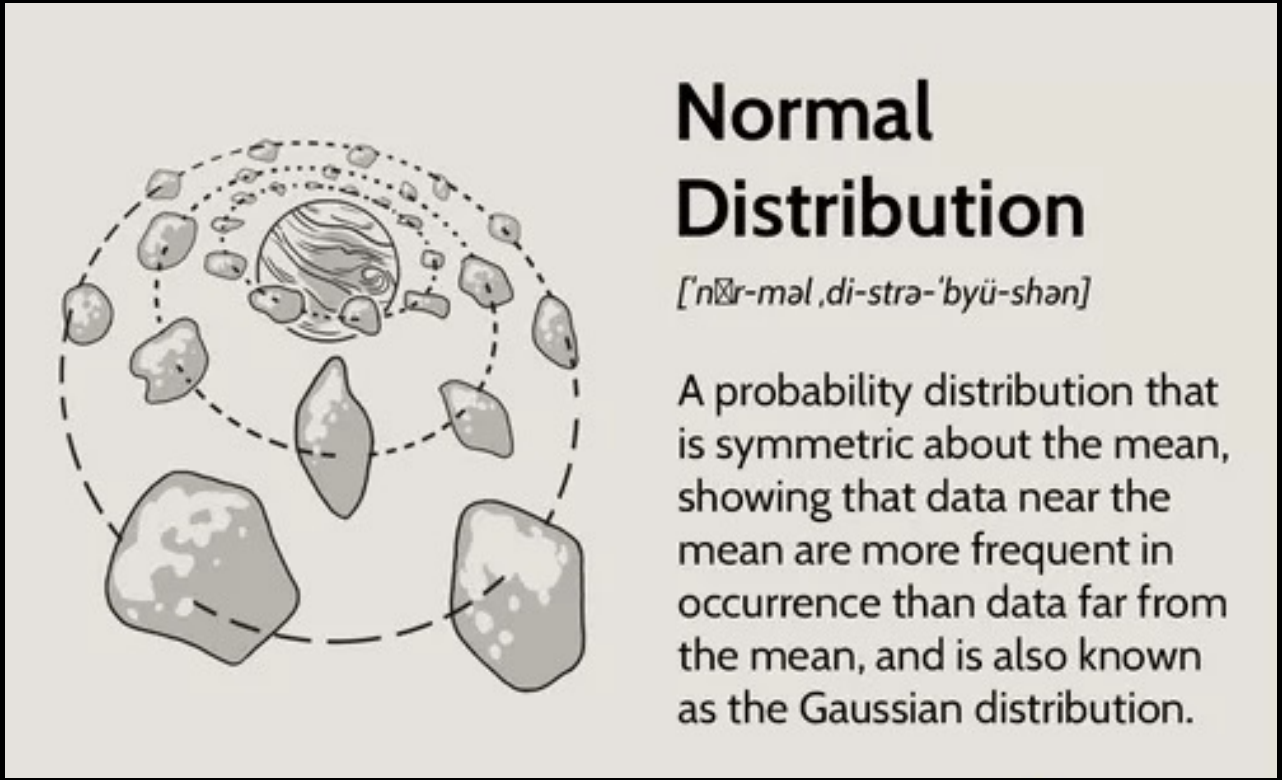
Because variance at first place will confirm that Whether the

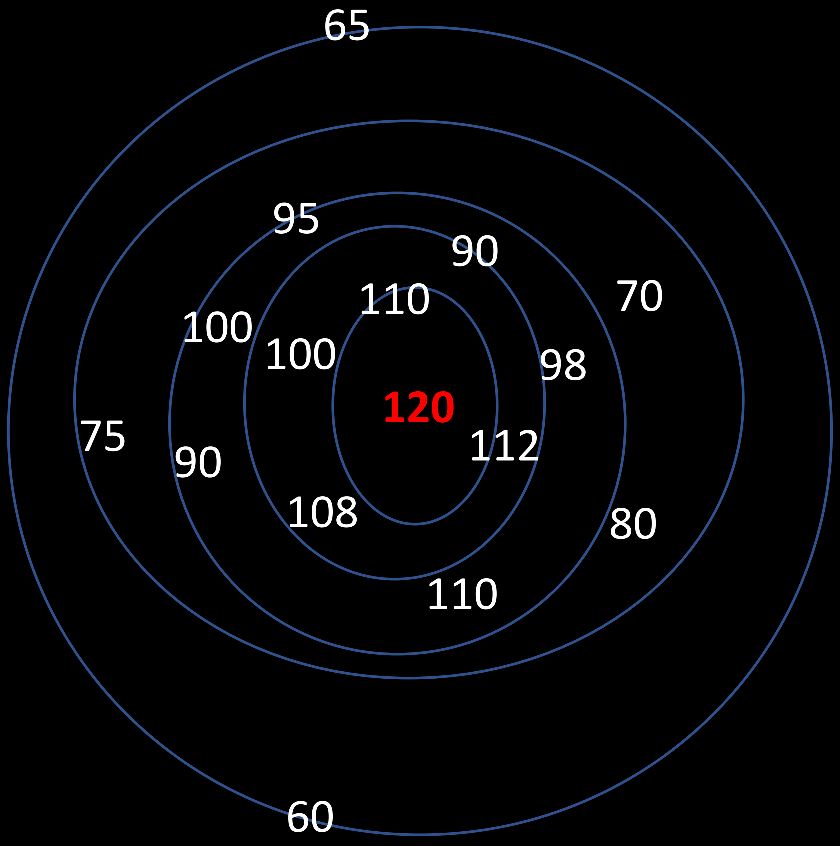
Moreover it will give the estimated range of difference .

Assumptions

1)Population from which samples are drawn is normal distribution .

What Is a Normal Distribution?





Assumptions

2) Samples are independent and random

Random and Independent Sample

Random sampling is a part of the sampling technique in which each sample has an equal probability of being chosen. A sample chosen randomly is meant to be an unbiased representation of the total population. Independent samples are samples that are selected randomly so that its observations do not depend on the values other observations.

Assumptions

3)Each one of the population has same variance.
4)Additivity of variance.

Hypothesis

μ = population mean

H0: All mean population should be equal

$\mu = \mu = \mu$

H1: At least one mean is different.

$\mu =! \mu =! \mu$

F-test in ANOVA

ANOVA uses the F-test to determine whether the variability between group means is larger than the variability of the observations within the groups. If that ratio is sufficiently large, you can conclude that not all the means are equal.

$$F_{(N,D)} = \frac{\text{Variation between Samples}}{\text{Variation within Sample.}}$$

Divide total Variation in two parts :

Variation between sample

variation b/w samples

$$X_1 \longleftrightarrow X_2 \longleftrightarrow X_3$$

X_1	X_2	X_3
20	18	25
21	20	28
23	17	22
16	25	28
20	15	32

$\Sigma X_1 = 100$ $\quad \Sigma X_2 = 95$ $\quad \Sigma X_3 = 135$

$\bar{X}_1 = 20$ $\quad \bar{X}_2 = 19$ $\quad \bar{X}_3 = 27$

Grand mean, $\bar{\bar{X}} = \dfrac{20+19+27}{3}$

$\bar{\bar{X}} = 22$

$H_0: \bar{X}_1 = \bar{X}_2 = \bar{X}_3$

$H_1: \bar{X}_1 \neq \bar{X}_2 \neq \bar{X}_3$

Variation within sample

$$X_1 \longleftrightarrow X_2 \longleftrightarrow X_3$$

X_1	X_2	X_3
20	18	25
21	20	28
23	17	22
16	25	28
20	15	32

Variation within Sample (error)

$\Sigma X_1 = 100$ $\quad \Sigma X_2 = 95$ $\quad \Sigma X_3 = 135$

$\bar{X}_1 = 20$ $\quad \bar{X}_2 = 19$ $\quad \bar{X}_3 = 27$

Grand mean, $\bar{\bar{X}} = \dfrac{20+19+27}{3}$

$\bar{\bar{X}} = 22$

$H_0: \bar{X}_1 = \bar{X}_2 = \bar{X}_3$

$H_1: \bar{X}_1 \neq \bar{X}_2 \neq \bar{X}_3$

one-way ANOVA

A one-way ANOVA uses one <u>independent variable</u>.

ANOVA tells you if the dependent variable changes according to the level of the independent variable. For example:

Example 1:

Independent variable is social media use

Levels :

-LOW

-MEDIUM

-HIGH

Dependent Variable

hours of sleep per night.

Example 2:

Independent variable is brand of soda

Levels:

-Coke

-Pepsi

-Sprite

Dependent Variable

price per 100ml.

Use a one-way ANOVA when you have collected data about one categorical independent variable and one quantitative dependent variable. The independent variable should have at least three levels (i.e. at least three different groups or categories).

Two way Anova

Two-way ANOVA uses two independent variable.

Independent variable is Fertilizer types

Levels 1, 2, and 3

Independent variable is Planting densities

Levels 1 and 2

Dependent Variable is

Crop growth measurement in tons.

Data and Sample Means

1)Calculate all means

S1	S2	S3	S4
12	7	5	5
9	10	8	8
12	13	11	8

Mean **11** **10** **8** **7**

Grand mean = 9

2)Sum of Squares of Error

(x-xbar)2

- **For the sample from population #1: $(12 - 11)^2 + (9 - 11)^2 + (12 - 11)^2 = 6$**
- **For the sample from population #2: $(7 - 10)^2 + (10 - 10)^2 + (13 - 10)^2 = 18$**
- **For the sample from population #3: $(5 - 8)^2 + (8 - 8)^2 + (11 - 8)^2 = 18$**
- **For the sample from population #4: $(5 - 7)^2 + (8 - 7)^2 + (8 - 7)^2 = 6$.** **48**

3)Sum of Squares of Treatment

Now we calculate the sum of squares of treatment.
Here we look at the squared deviations of each sample mean from the overall mean, and multiply this number by one less than the number of populations:4-1 = 3

(xbar-grand mean)2

$3[(11 – 9)^2 + (10 – 9)^2 +(8 – 9)^2 + (7 – 9)^2] = 3[4 + 1 + 1 + 4] =$ **30.**

4) Degrees of Freedom

- Variance between sample

Total sample – 1

4-1 = 3

- Variance within observations

Total number of observations –Total samples

12-4 =8

5.Mean Squares

- The mean square for treatment is 30 / 3 = **10.**
- The mean square for error is 48 / 8 = **6.**

6)The F-statistic

The final step of this is to divide the mean square for treatment by the mean square for error. This is the F-statistic from the data. Thus for our example $F = 10/6 = 5/3 = 1.667$.